STOR 565 · FALL 2026 · FINAL PROJECT

Additive faces: NMF versus PCA on the Olivetti faces

Can a face be built by adding patterns, and what does that cost compared with PCA?

Person 1 · Person 2 · Person 3 · Person 4

The question

DATA

Olivetti faces

400 photos: 40 people × 10 photos, 64 × 64 pixels, brightness in [0, 1]

BASELINE FROM CLASS

PCA

Signed directions around a mean face

NEW METHOD

NMF

A face is a sum of nonnegative patterns; nothing can be subtracted

What does the additive representation cost in accuracy, and what does it give in interpretability?

A fair comparison

- **Split:** 320 training and 80 test images, 8 + 2 per person, fixed
- **Fit on training images only;** both methods rebuild the same 80 test images
- **Error:** per-pixel mean squared error (MSE) on unclipped reconstructions
- **Same size:** the same number of components k (main case k = 40)
- **No labels used:** identities only check the split

CAVEAT

New photos of known people

Every person appears in both the training and the test images. We do not test unseen people.

The NMF model

$X \approx W H$, with $W \geq 0$ and $H \geq 0$

X: 320 × 4096

training images, one image per row

W: 320 × 40

how much of each pattern an image uses

H: 40 × 4096

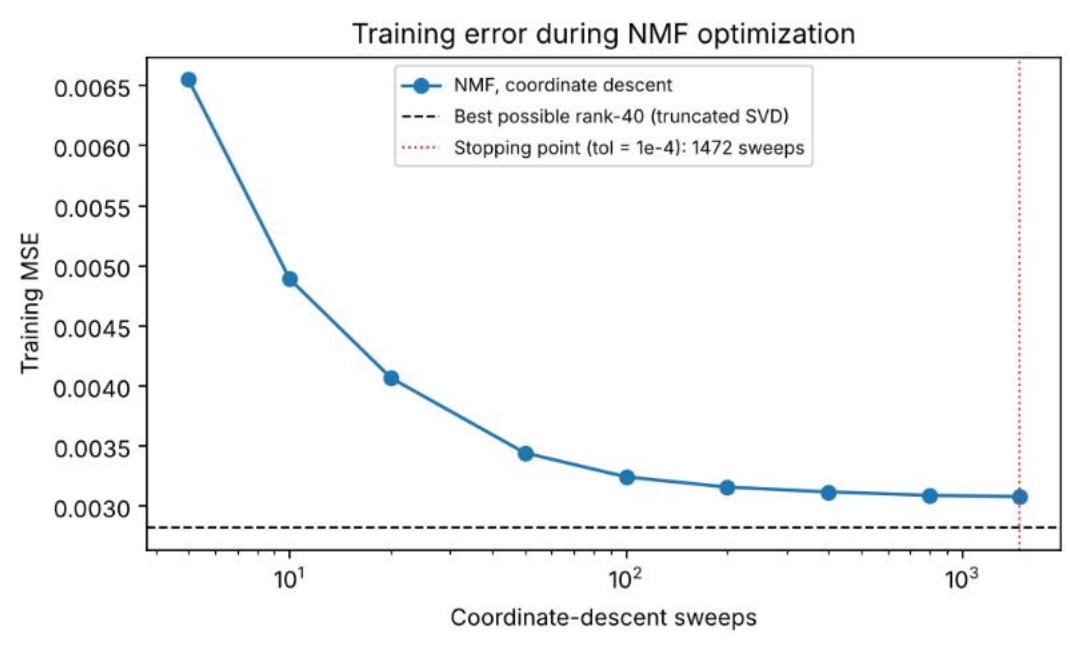
the shared patterns; each row is an image

One face = the sum, over the 40 patterns, of (weight × pattern). Nothing is subtracted. The 40 is a chosen size, not one pattern per person.

Objective: minimize $\frac{1}{2} \|X - W H\|^2$, the sum of squared pixel errors.

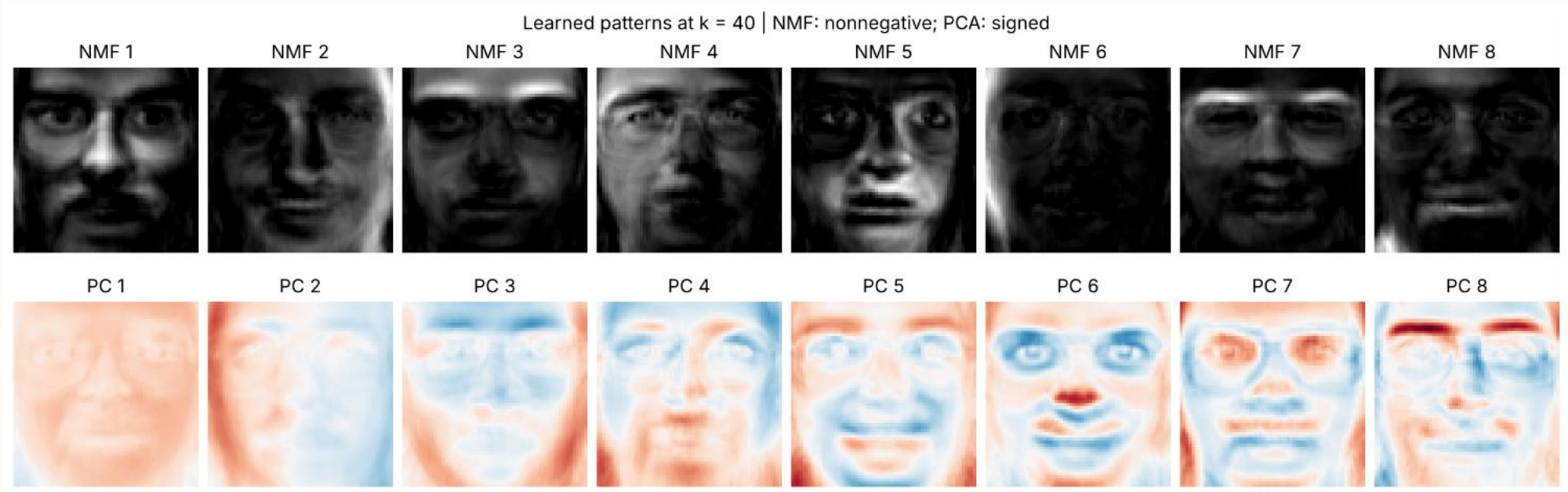
How the patterns are learned

- Convex in W for fixed H and in H for fixed W, but **not jointly convex**
- Coordinate descent: W[i, r] ← max(0, W[i, r] − gradient / a_r)
- Exact NMF is NP-hard: no guarantee of a global optimum
- A test face: keep H fixed and solve for nonnegative weights; an exact solver gives the same test MSE, 0.00394635



Training error falls fast, then slowly. It stays above the best possible rank-40 error (dashed).

What NMF learns: additive, but broad



0.317 vs 0.221

Hoyer sparsity, NMF patterns vs. PCA absolute loadings (0 = flat, 1 = one pixel)

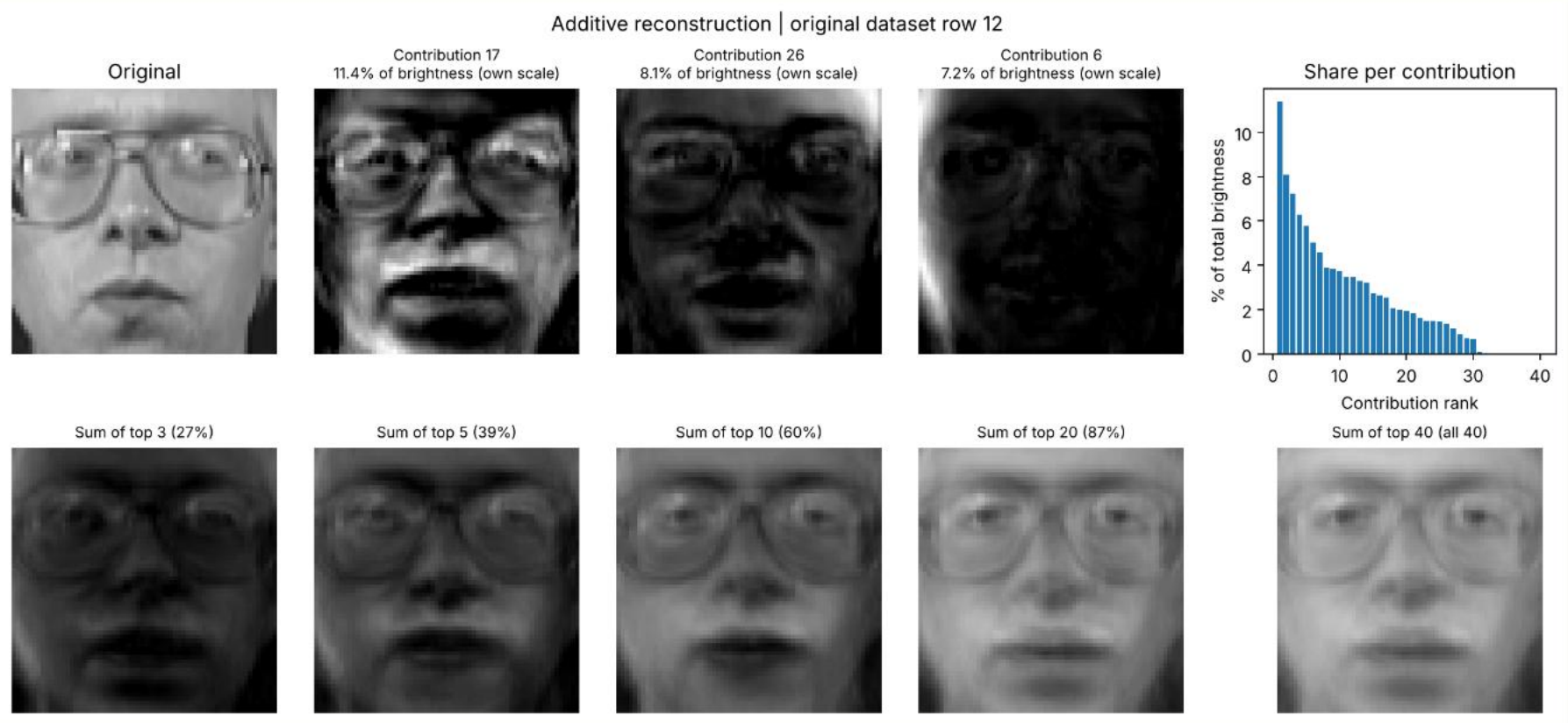
29.8 of 40

patterns used by a typical training image

Not parts

overlapping face-like patterns, not isolated eyes, nose or mouth

One face as a sum of contributions



Test image (dataset row 12): the largest of the 40 contributions supplies 11.4% of the brightness, and the top 20 supply 87%. No single pattern dominates.

Reconstruction at 40 components

Method	Test MSE	vs. PCA
PCA, 40 components + mean	0.003658	—
PCA, 39 components + mean	0.003696	+1.0%
Uncentered SVD, 40, no mean	0.003658	−0.005%
NMF, 40 components	**0.003946**	**+7.9%**

Not the mean term: PCA without a mean (uncentered SVD) is no better, and PCA with 39 components + mean, the same number of stored images as NMF, still wins.

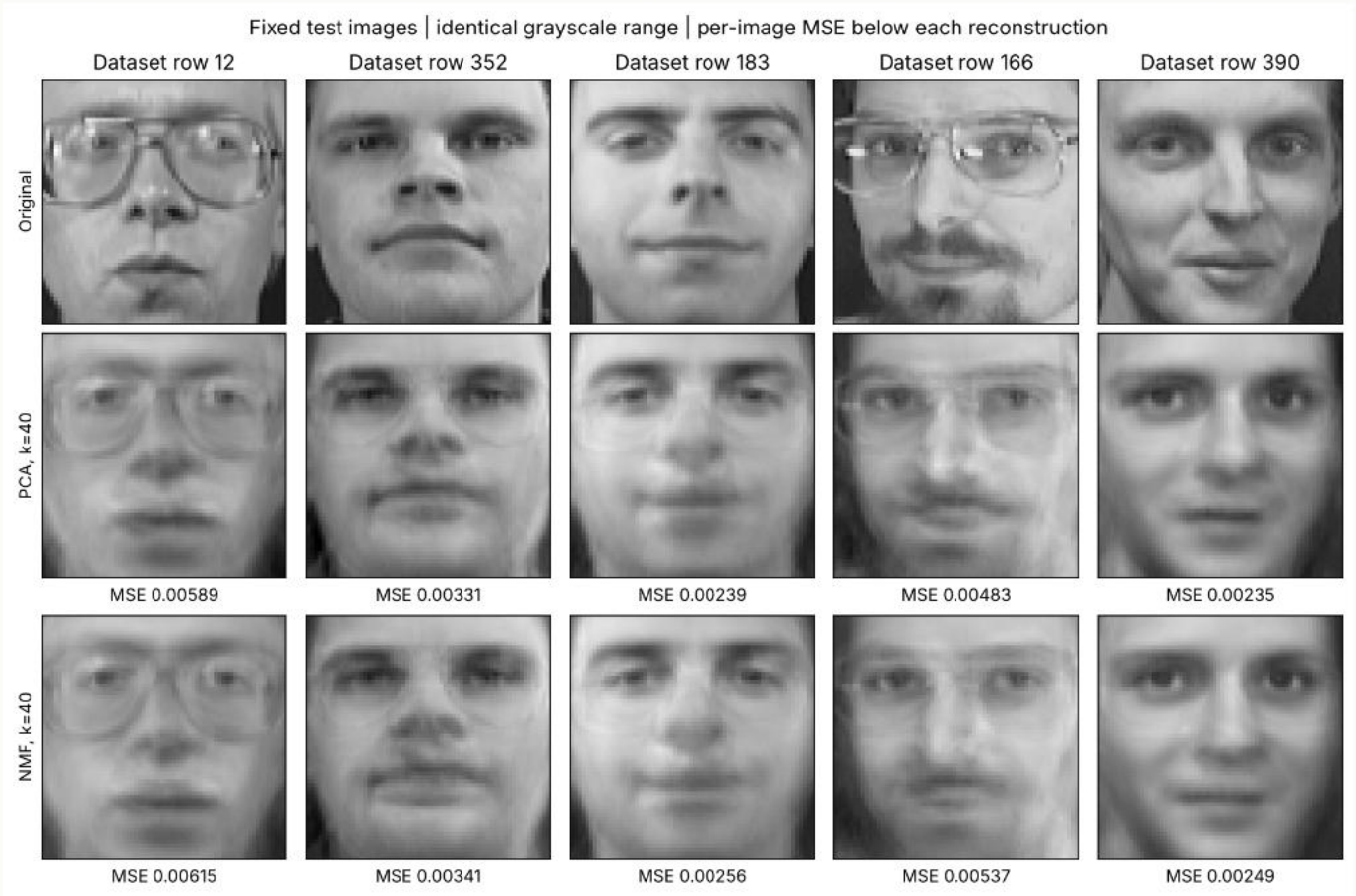
+7.9%

NMF test MSE above PCA

74 of 80

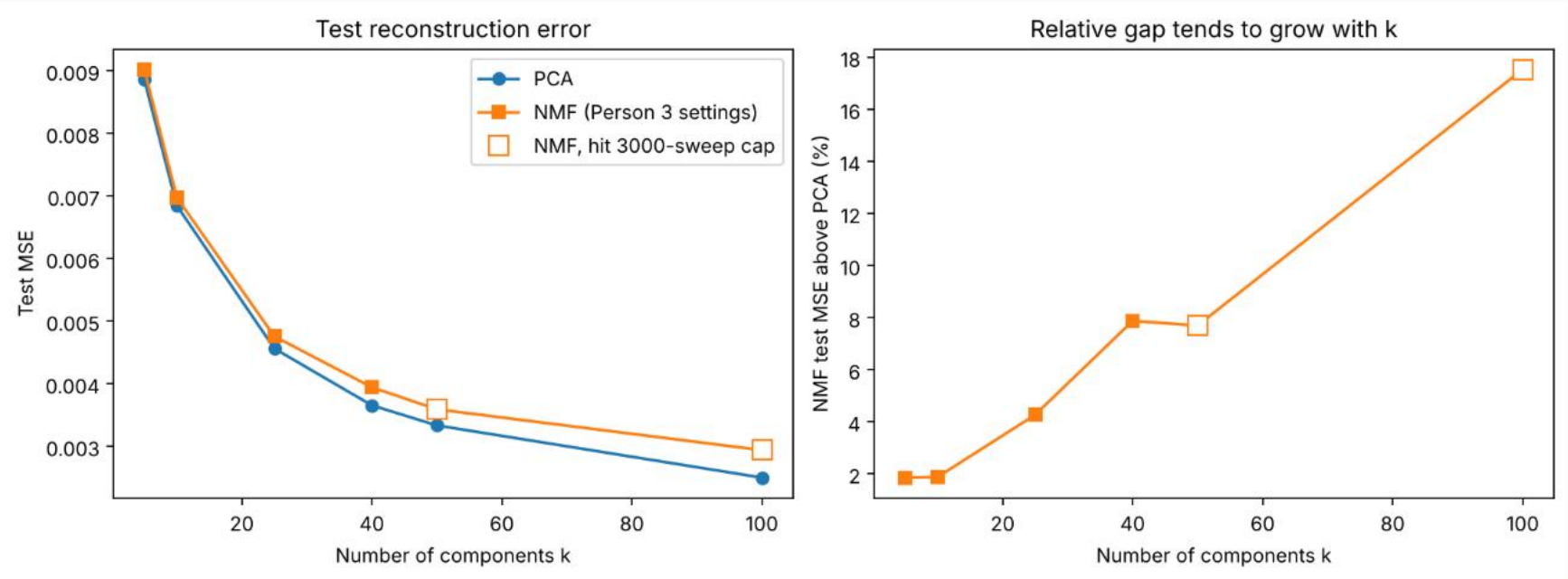
test images where PCA has the lower error

Same faces, two reconstructions



- Both blur fine detail and look very similar
- Largest gap on row 166, the face with glasses and a moustache: +11.2%
- A lower MSE means closer pixels, not better recognition

Is the gap robust?



+1.9% to +17.5%

NMF above PCA from k = 5 to k = 100; the two largest k hit the iteration cap, so those gaps are upper bounds

+5.5% to +7.9%

gap at k = 40 across initializations and tolerances; PCA is lower in every variant

Why is NMF worse?

Theory

By Eckart–Young, no rank-40 factorization $X \approx WH$ can beat the truncated SVD of the same training data.

NMF training MSE **0.003079** ≥ bound **0.002820**. The constraint costs accuracy.

Optimization

NMF finds a local solution, so part of the gap may come from the optimizer, not the model.

Other starts and stricter tolerances change the gap (5.5% to 7.9%) but never close it.

Gap = nonnegativity constraint + imperfect optimization, not the missing mean term.

What this does not show

- **One split.** The interval covers variation across the 80 test images, not across other splits or people.
- **Pixel error only.** It does not measure recognition, clustering or any downstream use.
- **One dataset and one objective.** 400 small, aligned faces; squared error only.
- **Local solutions.** Other optimizers or better initializations might shrink the gap, so we report a range.
- **Interpretability is partly visual.** Sparsity is measured; whether a pattern is a meaningful part is a judgment from looking at images.

Take-aways

1. NMF represents a face as a **sum of nonnegative patterns**; PCA cannot, because its components have mixed signs.
2. That constraint costs accuracy: NMF is **7.9% worse** at 40 components, and PCA is lower on 74 of 80 test images.
3. The gap is **not the mean term** and survives every variant we tried, so the trade-off is between fidelity and an additive, inspectable representation.

Code, data script and results: github.com/MachineWJQ6657/pca-vs-nmf

Backup: likely questions

Question	Short answer
Why is NMF not better at its own objective?	It is a rank-40 fit with a nonnegativity constraint, so its training error cannot be below the SVD bound (0.003079 vs. 0.002820).
Could better optimization close the gap?	Possibly part of it. Stricter tolerance and five random starts change the gap (5.5% to 7.9%) but PCA stays lower.
Is the missing mean the reason?	No. PCA without a mean (uncentered SVD) matches centered PCA, and PCA with 39 components + mean still beats NMF.
How are test faces encoded in NMF?	Patterns H are fixed; we solve a nonnegative least-squares problem for the weights. An exact solver gives the same test MSE.
What does the interval mean?	A paired bootstrap over the 80 test images (10,000 resamples). It does not cover other splits or people.